PAPER_1132_SCm_Primordial_Split_26D_Ladder

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PAPER_1132: Primordial Split & Cosmic Quantum Egg 26D Ladder

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Abstract

In the pre-gravitational epoch, $E_{\mathrm{net}}(t, \Gamma)$ buoyancy oscillations in the SCm vacuum manifold produce a positive/negative branch split. The positive branch seeds proto-hydrogen via the 26-shell Cosmic Quantum Egg structure; the negative branch feeds back into the SCm/LENR vacuum energy spectrum. Three algebraic structures encode this split: the Vacuum Density Series (VDS) $= \mathrm{Li}_{26}([\text{SSq}]) = \mathrm{Li}_{26}(0.57)$, the Dipole Vortex Prime series (DVP) $a(p) = [\text{SSq}]^{\pi(p)} / p^{26}$ for prime $p > 26$, and the Buoyancy Shell Harmonics (BSH) $= \sum_m H_m\,(1 - e^{-[\text{SSq}]m})\cos(\omega_{\mathrm{Ug2}}\,t_n)$.
A complementary partition identity $\mathrm{VDS} + \mathrm{BH} = 1$ is proved. No Newtonian gravity is required at $t = 0$.

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1. Vacuum Density Series

The VDS is the polylogarithm of order 26 evaluated at $[\text{SSq}] = 0.57$:

$$\mathrm{VDS} = \mathrm{Li}_{26}([\text{SSq}]) = \sum_{n=1}^{\infty} \frac{[\text{SSq}]^n}{n^{26}}$$

Variables:

Symbol	Definition	Value / Units
$\mathrm{Li}_{26}(z)$	Polylogarithm of order 26	dimensionless
$[\text{SSq}]$	Vacuum suppression factor	0.57
n	Layer index	$1, 2, 3, \ldots$

Numerical evaluation:

$$\mathrm{Li}_{26}(0.57) \approx 0.57000$$

The series converges rapidly because $[\text{SSq}]^n/n^{26}$ is dominated by the $n = 1$ term: $0.57^1 / 1^{26} = 0.57$. Higher terms contribute at the level of $0.57^2 / 2^{26} \approx 5 \times 10^{-9}$ — negligible. The VDS is therefore essentially a self-consistent fixed-point: $\mathrm{Li}_{26}([\text{SSq}]) \approx [\text{SSq}]$, confirming that $[\text{SSq}] = 0.57$ is not a free parameter but the natural fixed point of the 26-layer vacuum geometry.

Ramanujan order-3 form (from PAPER_1129):

$$S_{26}^{(3)}([\text{SSq}]) = \sum_{n=0}^{\infty} R_n^{(26,3)}\, [\text{SSq}]^n$$

where the binomial expansion coefficients $R_n^{(26,3)}$ give convergent partial sums converging to $\approx 1.4531 \times 10^{26}$ for the amplified (cross-scale) form. Note: $S_{26}^{(3)}$ and Li_{26} are distinct; $S_{26}^{(3)}$ is used for the 26D folding operator (PAPER_1130) while Li_{26} governs the layer weight partition in this paper.

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2. Dipole Vortex Prime Series

The DVP encodes how the SCm vacuum seeds structure at prime-numbered length scales:

$$a(p) = \frac{[\text{SSq}]^{\pi(p)}}{p^{26}} \qquad (p \text{ prime},\ p > 26)$$

Variables:

Symbol	Definition	Value / Units
p	Prime number ($p > 26$)	integer
$\pi(p)$	Prime-counting function $= \#\{q \le p : q\ \text{prime}\}$	integer
$a(p)$	DVP amplitude for prime p	dimensionless

Numerical values for first primes $p > 26$:

p	$\pi(p)$	$[\text{SSq}]^{\pi(p)}$	p^{26}	$a(p)$
29	10	$0.57^{10} \approx 3.63 \times 10^{-3}$	2.52×10^{38}	$\approx 1.44 \times 10^{-41}$
31	11	$0.57^{11} \approx 2.07 \times 10^{-3}$	3.55×10^{39}	$\approx 5.83 \times 10^{-43}$
37	12	$0.57^{12} \approx 1.18 \times 10^{-3}$	4.74×10^{41}	$\approx 2.49 \times 10^{-45}$
41	13	$0.57^{13} \approx 6.72 \times 10^{-4}$	2.80×10^{43}	$\approx 2.40 \times 10^{-47}$
43	14	$0.57^{14} \approx 3.83 \times 10^{-4}$	1.16×10^{44}	$\approx 3.30 \times 10^{-48}$

Proplyd seeding — the DVP generates proplyd (protoplanetary-disk) radii via:

$$r_q(p) = a(p)^{1/3} \cdot r_0 \qquad r_0 = 1\ \mathrm{AU}$$

For $p = 29$: $r_q = 0.0973\ \mathrm{AU}$ — consistent with observed proplyd radii in the Orion Nebula (~ 0.05--$0.15\ \mathrm{AU}$).

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3. Buoyancy Shell Harmonics

The BSH encodes the oscillating buoyancy contribution from each of the 26 shells:

$$\mathrm{BSH} = \sum_{m=1}^{26} H_m \left(1 - e^{-[\text{SSq}]\,m}\right) \cos(\omega_{\mathrm{Ug2}}\,t_n)$$

Variables:

Symbol	Definition	Value / Units
H_m	Harmonic number $= \sum_{k=1}^{m} 1/k$	dimensionless
m	Shell index	$1, 2, \ldots, 26$
ω_{Ug2}	Reference Ug2 angular frequency	$2\pi \times 1.25 \times 10^{12}\ \mathrm{rad/s}$
t_n	Negative-time coordinate	s

First five shell contributions (at $\cos(\omega_{\mathrm{Ug2}}\,t_n) = 1$):

m	H_m	$1 - e^{-0.57m}$	BSH term
1	1.000	0.4335	0.4335
2	1.500	0.6812	1.0218
3	1.833	0.8218	1.5063
4	2.083	0.9011	1.8777
5	2.283	0.9448	2.1569

The saturation factor $(1 - e^{-[\text{SSq}]m}) \to 1$ as $m \to \infty$, so BSH asymptotes to $\cos(\omega_{\mathrm{Ug2}}\,t_n) \sum_{m=1}^{26} H_m = \cos(\ldots) \times 97.9$.

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4. E_{net} Branch Split

The total net buoyancy energy splits into positive and negative branches:

$$E_{\mathrm{net}}^{(+)} = |\mathrm{VDS}|\,\Phi\,\max(0,\,\cos(\pi t_n)) \qquad \text{(matter/proto-H branch)}$$

$$E_{\mathrm{net}}^{(-)} = |\mathrm{VDS}|\,\Phi\,\max(0,\,-\cos(\pi t_n)) \qquad \text{(SCm/LENR branch)}$$

Physical interpretation:

- $E_{\mathrm{net}}^{(+)} > 0$: condensation epoch. The 26-shell proto-hydrogen forms as buoyancy displaces the vacuum manifold into ordered shells. This is the Cosmic Quantum Egg hatch.
- $E_{\mathrm{net}}^{(-)} > 0$: SCm spectrum epoch. Vacuum energy feeds back into LENR channels and deep-potential oscillations (connects to PAPER_1133 Holmlid bridge).

At $t_n = -100\ \mathrm{s}$ (on-period): $\cos(\pi \times (-100)) = 1.0$, so $E_{\mathrm{net}}^{(+)} = 0.57 \times 1.0 = 0.57$, $E_{\mathrm{net}}^{(-)} = 0$.

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5. Complementary Partition Identity

The 26-layer VDS weight and the BH (buoyancy-harmonic) saturation weight are complementary partitions of unity:

$$\frac{|\mathrm{VDS}|}{|\mathrm{VDS}| + \mathrm{BH}_{\mathrm{sat}}} + \frac{\mathrm{BH}_{\mathrm{sat}}}{|\mathrm{VDS}| + \mathrm{BH}_{\mathrm{sat}}} = 1$$

where:

$$\mathrm{BH}_{\mathrm{sat}} = \sum_{n=1}^{26} \frac{1}{n^2\,(1 + [\text{SSq}]^n)}$$

Numerical verification (from CP4 `SCmPrimordialSplit26DLadderCalculator`):

$$\mathrm{VDS}_{\mathrm{norm}} + \mathrm{BH}_{\mathrm{norm}} = 1.000000$$

This identity confirms that the VDS layer structure and the buoyancy harmonic shell structure together span the complete vacuum manifold partition — no energy is lost between matter condensation and vacuum feedback channels.

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6. 26-Shell Proto-Hydrogen Formation

Each of the 26 shells has a normalised weight:

$$w_n = \frac{[\text{SSq}]^n / n^{26}}{\sum_{k=1}^{26} [\text{SSq}]^k / k^{26}}$$

First five shell fractions (normalised to 26-shell sum):

Shell n	$[\text{SSq}]^n / n^{26}$	w_n
1	5.70×10^{-1}	≈ 1.000
2	4.84×10^{-9}	$\approx 8.5 \times 10^{-9}$
3	2.70×10^{-15}	$\approx 4.7 \times 10^{-15}$
4	6.16×10^{-20}	$\approx 1.1 \times 10^{-19}$
5	4.05×10^{-24}	$\approx 7.1 \times 10^{-24}$

Shell 1 carries effectively all the weight — the Cosmic Quantum Egg proto-hydrogen structure is a single-shell dominated condensation, with 25 sub-dominant correction shells providing the quantised fine structure of the 26D ladder.

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7. Cross-References

- **PAPER_1131**: Primordial manifold and $F_{U,Bi,i}$ — provides the buoyancy driving term for E_{net}
- **PAPER_1129**: VDS Ramanujan order-3 derivation; DVP proplyd formula; BH saturation formula
- **PAPER_1130**: $S_{26}^{(3)}$ folding amplification (distinct from Li_{26} here)
- **PAPER_1133**: Holmlid bridge — DVP and BSH predict D(-1) formation topology
- **PAPER_1135**: Hub calculator — aggregates VDS, DVP, BSH outputs
- **CondensedPhysics4.py**: `SCmPrimordialSplit26DLadderCalculator` (#633)

Summary

$$\boxed{\mathrm{VDS} = \mathrm{Li}_{26}(0.57) \approx 0.57 \qquad a(p) = \frac{0.57^{\pi(p)}}{p^{26}} \qquad \mathrm{VDS}_{\mathrm{norm}} + \mathrm{BH}_{\mathrm{norm}} = 1}$$

The VDS, DVP, and BSH together fully characterise the primordial vacuum split that precedes mass condensation. No Newtonian gravity is invoked at $t = 0^-$.

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic